

PR&R — Polymarket Real-Time Monitor

A live market monitor and wallet tracker for Polymarket's up/down cycle markets. Stitches together Binance spot/futures, Chainlink, and Polymarket price feeds into one synchronized chart, plots any tracked wallet's trades as they happen, and can log every cycle to CSV for later analysis.

1. Requirements

What	Version
Node.js	22 or newer (tested on 22.16)
npm	10 or newer (ships with Node 22)
macOS / Linux / Windows	Any (macOS recommended — some file-picker features use the system dialog)
Disk space	~500 MB for dependencies + whatever you want for CSV logs
Network	Outbound WebSocket + HTTPS to polymarket.com , binance.com , chain.link

No database required. No API keys required to get started — the app works out of the box using Polymarket's public feeds.

2. Installation

```
# 1. Clone / copy the project to your machine
cd /path/to/LEECHTRACKER

# 2. Install dependencies
npm install
```

The install pulls Next.js 15, React 19, and a few WebSocket clients. First run takes 1–2 minutes; subsequent runs are instant.

3. Running

```
npm run dev
```

This starts the dev server on <http://localhost:3030>. Open that URL in any modern browser (Chrome, Safari, Firefox, Edge).

When you're done, press **Ctrl+C** in the terminal to stop.

If `npm run dev` does nothing

The most common cause is a leftover Next.js process holding port 3030. Check and kill it:

```
lsof -tiTCP:3030 -sTCP:LISTEN | xargs kill  
npm run dev
```

If you see `ENOENT: uv_cwd`

The terminal's cached working directory is stale (usually after a volume unmount/remount). Just `cd` back into the project directory:

```
cd /path/to/LEECHTRACKER
```

4. First-Time Setup (in the UI)

4.1 Settings page

Navigate to **Settings** in the top nav. There are two groups:

Chainlink Data Streams (*optional*) Fills in a Chainlink-backed BTC price card on the Indicators page. If you don't care about the Chainlink oracle price alongside Binance, skip this.

- **Chainlink API Key** — from chain.link/data-streams
- **Chainlink User Secret** — paired signing secret

Endpoints (*optional*) Override the default Polymarket URLs if you're routing through a proxy. Leave blank to use Polymarket's public endpoints.

Click **Save changes** after entering any keys.

4.2 Tracked Wallets

Still on the Settings page, scroll down to **Tracked Wallets**.

1. Paste a Polymarket trader's wallet address (`0x...`)
2. Give them a label (e.g. "Degen Mike")
3. Click **Add wallet**

You can track up to **10 wallets**. Use the **On / Off** toggle on each wallet to choose which trader's activity is plotted on the chart (only one is active at a time).

4.3 Log Output (*only needed if you plan to log cycles*)

At the bottom of Settings, set **Log Output** to a folder where the cycle logger should write CSVs. Pick any folder **outside** the project directory (Next.js hot-reloads if you write inside the source tree).

Good defaults:

- macOS: `/Users/you/Documents/PR-R-logs`
- Windows: `C:\Users\you\Documents\PR-R-logs`

You can also click **Browse** to open a system folder picker.

5. Using the App

Indicators page (home)

This is the live monitor. You'll see:

- **Status pills** at the top — green when each feed is connected.
- **Price cards** — Binance Spot, Binance Futures, Chainlink, UP/DOWN share prices.
- **Market selector** — pick the asset (BTC / ETH / SOL / XRP) and timeframe (5m / 15m).
- **Cycle info** — current cycle slug, spot/futures spread, progress bar.
- **Chart** — the live overlay. X-axis is the cycle; Y-axis is price.
 - Blue = Binance Spot, Violet = Futures, Amber = Chainlink.
 - Green = UP share price, Red = DOWN share price.
 - Orange/pink markers = trades from the active tracked wallet.
 - Scroll to zoom the price range.
- **Orderbooks** — 10 levels of UP and DOWN book, updated in real time.

Market Log page

A live feed of every new trade tick on the currently selected market — useful for watching order flow without the chart noise.

Cycle Logger (on Indicators)

The **Cycle Logger** panel controls CSV logging. Before starting:

1. Make sure at least one wallet is tracked.
2. Make sure a log folder is set (Settings → Log Output).

Click **Start logger**. While running, every 5-minute cycle produces three files:

- `orderbooks/{cycle}/orderbook.csv` — 300 rows, full order book every second.
- `prices/{cycle}/prices.csv` — 300 rows, prices only (slim version).
- `{wallet-label}/{cycle}/traderactivity.csv` — the tracked wallet's trades alongside per-second prices.

Click **Stop logger** to halt. Files are flushed to disk on stop.

6. How trade tracking works (no API key needed)

The app subscribes to Polymarket's **Real-Time Data Socket** (free, public, no authentication) and filters every trade by the wallets you've configured. This means:

- Add a wallet → its trades appear on the chart within a second of hitting the chain.
- You can track anyone — no permission needed from the wallet owner.
- Works entirely through the browser and localhost, no external services.

7. Troubleshooting

Symptom	Fix
Links in the top nav don't navigate	Hard-reload the page (Cmd+Shift+R / Ctrl+Shift+R). Usually caused by a stale dev bundle after code changes.
<code>npm run dev</code> exits immediately with <code>EADDRINUSE</code>	Port 3030 is held by a zombie process. Run <code>lsof -tiTCP:3030 -sTCP:LISTEN xargs kill</code> .
<code>ENOENT: uv_cwd</code> in terminal	Shell's cached working directory is stale. <code>cd</code> back into the project.

"Wallet Stream" pill red	No tracked wallet is toggled on. Go to Settings and toggle one.
Chart shows "Waiting for next cycle"	Polymarket's upcoming cycle slug hasn't been published yet; it'll populate within ~5 seconds.
Cycle logger won't start	Check that (a) at least one wallet is tracked and (b) a Log Output folder is set.

8. File layout (for the curious)

```
src/
  app/
    indicators/      - live monitor page (the home page)
    market-log/     - live trade feed page
    settings/        - credentials + wallets + log-path UI
    api/
      crypto/        - Binance / Chainlink / 5m-markets endpoints
      polymarket/    - CLOB book fetcher
      wallet-tracker/
        trades-stream/ - Polymarket RTDS bridge (SSE → browser)
      cycle-logger/   - start/stop/config endpoints for the logger
      settings/       - read/write credentials
    components/
      indicators/     - chart + cycle-logger panel + market selector
      nav/            - top navigation
    hooks/            - Binance / Chainlink / wallet-stream / book WS
    lib/
      cycleLogger/    - the CSV-writing service
      settings.ts     - credentials schema + storage
    public/
      logo.png        - the brand logo in the nav
```

User data is stored in `~/.leechtracker/` :

- `config/settings.json` — your saved API keys
- `config/config.json` — tracked wallets + log path

Nothing is synced to a remote server. Everything runs locally.

9. Building for production

```
npm run build
npm run start
```

Production server runs on port 3030 by default. Set `PORT=4000` in front of the command to change.

10. Getting help

- Check the **Troubleshooting** table above first.

- Open the browser DevTools Console — most client-side errors surface there with a useful stack trace.
- Check the terminal running `npm run dev` — server-side errors print there.